Reinforcement Learning & Bayesian Methods for Cryptocurrency Portfolio Management

DQN · REINFORCE · Market Simulation · Bayesian Bandits

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Slide 1 Introduction & Professional Background

About Me

- Graduate student, Cornell University
  ORIE — Operations Research & Information Engineering
- Research focus: reinforcement learning, financial optimization, Bayesian inference
- Prior work: quantitative modeling in financial time-series and sequential decisions

This Project

- 4-part study applying RL and Bayesian methods to live cryptocurrency markets
- Asset universe: BTCB, ETH, SOL, BNB, USDC
- Combines *trading policy design* with *systemic / regulatory market analysis*

Four-Part Framework

Part 1	Environment design & feature engineering
Part 2	Deep Q-Network (DQN) trading agent
Part 3	REINFORCE + 3 trader archetypes, Shapley attribution, market simulation
Part 4	Bayesian bandits: UCB1, Bayes-UCB, Thompson Sampling on 5 crypto arms

Why Crypto?

Extreme regime shifts, manipulative microstructure, and herd dynamics make crypto an ideal stress-test for adaptive RL and sequential Bayesian decisions.

Slide 2 Benefits to the End User

Institutional Asset Manager (BlackRock-like)

- Capital preservation: REINFORCE-Manipulator returned −6% vs. −46% buy-and-hold — preserved 94% of capital in bear market
- Explainable AI: Shapley values identify exact signals driving each policy (RSI for arb; volume for manipulator; 7-day return for herder) — auditable to compliance
- Regime-aware allocation: policies adapt dynamically to rise / steady / decline
- Bayesian portfolio selection: Thompson Sampling allocates across 5 assets with quantified posterior uncertainty, $\Gamma_T = 0.15$

Regulator / Government Body (DOB)

- Manipulator detection: front-runners identified via microstructure (MA deviation, realized vol, volume) — not price level; requires *order-book* surveillance
- Crisis early warning: herder-dominated markets generate 6.75× more cascade events than arbitrageur markets
- Stablecoin surveillance: arbitrageur surge predicts USDT peg stress (grounded in actual USDT-USD hourly deviation data)
- Equal-mix policy: 33/33/34 trader mix achieves best balance of efficiency, stability, and crisis frequency

The same model serves both trading desks and regulatory surveillance.

Slide 3 Data Description

Primary Dataset — BTCB-USD

Asset	BTCB-USD (Bitcoin BEP-20)
Frequency	Hourly OHLCV
Observations	17,248 hourly rows
Period	Mar 2024 – Feb 2026
Train split	Mar 2024 – Oct 2025
Test split	Oct 2025 – Feb 2026

Key Periods

Training (Mar 2024 – Oct 2025):
≈13,000 hourly obs.; varied regimes

Test (Oct 2025 – Feb 2026):
BTCB declined 46.2% — severe bear
Critical stress-test for all agents

Bandit horizon ($T = 500$):
Bootstrap resampled; 100 replications

Supplementary Data

Cross-assets	ETH, Gold, Silver, S&P500 (correlation study by regime)
Stablecoin	USDT-USD hourly peg data

Market Regimes & ETH Correlation

Regime	% Hours	ETH Corr
Steady	64%	0.756
Decline	21%	**0.810**
Rise	15%	0.677

RL Agents (Parts 2 & 3)

DQN — Part 2

Off-policy Q-learning; ε-greedy; replay buffer
Actions: $\{$Buy, Hold, Sell$\}$; 13-dim state vector

REINFORCE + Baseline — Part 3

On-policy Monte Carlo PG; 2-layer MLP (128 units)
$\nabla_\theta J = \mathbb{E}[\log \pi(a|s;\theta) \cdot (G_t - V(s))]$

Reward-shaped trader archetypes:

Arb: $r - 0.4 \cdot \sigma_{24} \cdot f_{\text{crypto}}$
Manip: $r + 0.15 \cdot \bar{V} \cdot \text{sign}(r_{4h}) \cdot d$
Herder: $r + 0.25 \cdot \text{sign}(r_{24h}) \cdot d$

Explainability: SHAP KernelExplainer on $\pi(\text{Buy}|s)$

Bayesian Bandits (Part 4)

Setup

$K = 5$ arms; $Y_t = \mathbf{1}[\text{8h return} > 0]$; Beta(3,3) prior; $T = 500$

Four Algorithms

UCB1	$\hat{\mu}_k + \sqrt{2 \ln t / n_k}$
Bayes-UCB	$Q(1 - 1/t,\ \text{Beta}(\alpha_k, \beta_k))$
Thompson	$\hat{\theta}_k \sim \text{Beta}(\alpha_k, \beta_k)$
Greedy	$\arg\max_k\ \alpha_k/(\alpha_k + \beta_k)$

Info ratio: $\Gamma_t = [\mathbb{E}[\Delta_t]]^2 / I_t(\tilde{\theta}; A_t)$

Market Simulation (Part 3)

7 trader compositions; 500 test-set steps
Signal: $S_t = w_{\text{arb}} d_{\text{arb}} + w_{\text{man}} d_{\text{man}} + w_{\text{herd}} d_{\text{herd}}$

Trading Performance

- REINFORCE-Manipulator: -6% vs. -46% buy-and-hold; preserved 94% of capital
- DQN collapsed to constant HOLD (Q-values: Hold=0.0043, Buy=0.0041, Sell=0.0033)
  Stochastic RL adapts; deterministic RL does not
- Bayesian: Thompson Sampling regret 9.19 ± 0.78;
  info ratio $\Gamma_T = 0.149 \ll 0.5$ bound
  Optimal arm: BNB ($\theta^* = 0.519$)

Shapley Attribution

- **Arb:** RSI + 24h return (mean reversion)
- **Manipulator:** MA deviation + realized vol (pure microstructure — zero trend signals)
- **Herder:** 7-day return dominant;
  blind to RSI, volume, contrarian signals

Systemic / Regulatory Findings

- Herder markets: $6.75\times$ more crisis events than arbitrageur markets (0.054 vs. 0.008)
- All-manipulator markets: 0 measured crises but wealth extraction is *silent*
  $\Rightarrow$ requires order-book surveillance
- Arbitrageur surge predicts USDT peg stress (lowest stability: 18.1 vs. 22.6 all-manipulator)
- Equal mix (33/33/34) is the regulatory sweet spot:
  balanced crisis, stability, and efficiency

Core Takeaway

Stochastic RL dominates deterministic Q-learning in volatile regimes. Manipulator activity is best detected via microstructure — not price triggers. A diverse trader ecosystem minimizes systemic risk.